

Algebra: Essential Factoring Techniques

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DIRECTIONS

Factor each expression using GCF, difference of squares, sum/difference of cubes, or trinomials

1 Factor (sum of cubes):

$$x^3 + 27$$

Answer: _____

2 Factor (sum of cubes):

$$x^3 + 8$$

Answer: _____

3 Factor (difference of squares):

$$x^2 - 1$$

Answer: _____

4 Factor the trinomial:

$$x^2 + 8x + 15$$

Answer: _____

5 Factor the trinomial:

$$x^2 - x - 2$$

Answer: _____

6 Factor out the GCF:

$$4x^2 + 4x$$

Answer: _____

7 Factor (difference of cubes):

$$x^3 - 27$$

Answer: _____

8 Factor out the GCF:

$$20x^2 + 20x$$

Answer: _____

9 Factor (difference of squares):

$$x^2 - 1$$

Answer: _____

10 Factor out the GCF:

$$8x^2 + 12x$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Always factor out GCF first. Then check: diff of squares? Sum/diff of cubes? Trinomial?

1 Factor (sum of cubes):

$$= (x + 3)(x^2 - 3x + 9)$$

$$x^3 + 27$$

2 Factor (sum of cubes):

$$= (x + 2)(x^2 - 2x + 4)$$

$$x^3 + 8$$

3 Factor (difference of squares):

$$= (x - 1)(x + 1)$$

$$x^2 - 1$$

4 Factor the trinomial:

$$= (x - -5)(x - -3)$$

$$x^2 + 8x + 15$$

5 Factor the trinomial:

$$= (x - -1)(x - 2)$$

$$x^2 - x - 2$$

6 Factor out the GCF:

$$= 2x(2x + 2)$$

$$4x^2 + 4x$$

7 Factor (difference of cubes):

$$= (x - 3)(x^2 + 3x + 9)$$

$$x^3 - 27$$

8 Factor out the GCF:

$$= 4x(5x + 5)$$

$$20x^2 + 20x$$

9 Factor (difference of squares):

$$= (x - 1)(x + 1)$$

$$x^2 - 1$$

10 Factor out the GCF:

$$= 4x(2x + 3)$$

$$8x^2 + 12x$$



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